Sagnik Chatterjee

sagnik.chatterjee@tifr.res.in chatsagnik@gmail.com chatsagnik.github.io

Education and Employment

Dec 2025 – Present	**Visiting Fellow (Postdoctoral Researcher)** at STCS, TIFR Mumbai. **Host:** Prof. Jatin Batra.
Jul 2019 – Jun 2025	**Ph.D.** in Computer Science and Engineering at IIIT-Delhi. **Advisor:** Prof. Debajyoti Bera.
Sep 2017 – Mar 2019	**Staff Consultant,** Oracle Financial Services Software Limited, Bangalore.
Aug 2013 – Aug 2017	**B.Tech.** in Computer Science and Engineering at MAKAUT (formerly, WBUT-Inhouse). **GPA:** *8.10/10.*

Research

Ph.D. Thesis

1. **S. Chatterjee**, "Designing quantum learning algorithms for classical objects," Available at https://repository.iiitd.edu.in/xmlui/handle/123456789/1758, PhD thesis, IIIT-Delhi, May 2025.

Conference Proceedings

1. S. Das†, **S. Chatterjee**†, and J. Batra, "Convex basins in single-index model loss landscapes: Applications to robust recovery under strong adversarial corruption," in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.
2. M. Mukherjee, **S. Chatterjee**, and A. Sethi, "Perfect secret key generation for a class of hypergraphical sources," in *IEEE International Symposium on Information Theory (ISIT)*, 2026. URL: https://arxiv.org/abs/2601.10697.
3. **S. Chatterjee**†, M. Mukherjee†, and A. Sethi†, "Generalization bounds for dependent data using online-to-batch conversion," in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. URL: https://proceedings.mlr.press/v258/chatterjee25b.html.
4. **S. Chatterjee**, T. SAPV, and D. Bera, "Efficient quantum agnostic improper learning of decision trees," in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024. URL: https://proceedings.mlr.press/v238/chatterjee24a.html.

† denotes equal contribution.

Journal Articles

1. **S. Chatterjee**, R. Bhatia, P. S. Chani, and D. Bera, "Quantum boosting using domain-partitioning hypotheses," *Quantum Machine Intelligence*, vol. 5, no. 2, pp. 1–20, 2023. DOI: 10.1007/S42484-023-00122-3.

Under Review

1. **S. Chatterjee**, "The Quantum Learning Menagerie (A survey on Quantum learning for Classical concepts)," 2026. URL: https://arxiv.org/abs/2602.01054.
2. Y. Saxena, T. Sapv, **S. Chatterjee**, and R. K. Lee, "Generalized cross-resonance scheme for maximally-entangling two-qutrit gates," 2026. URL: https://arxiv.org/abs/2504.15265.

† denotes equal contribution.

Workshop Papers and Posters

1. **S. Chatterjee** and V. Kungurtsev, *Quantum solutions to the privacy vs. utility tradeoff*, 2023. arXiv: 2307.03118.
2. **S. Chatterjee** and D. Bera, *Applying the quantum alternating operator ansatz to the graph matching problem*, Extended Abstract at the The 20th Asian Quantum Information Science (AQIS) Conference, 2020. arXiv: 2011.11918.

Teaching

Instructor (Short Courses)

Mar 2026 — Designed and taught a mini-course (6 lectures) at **IIT (ISM) Dhanbad**.
Topic: axioms of quantum computing, quantum query speedups for promise problems, variational quantum algorithms, and hands-on introduction to QISKIT.

Feb 2024 — Designed and taught 3 lectures at the **Chennai Mathematical Institute**.
Topic 1: The Harrow-Hassidim-Lloyd Algorithm and its extensions.
Topic 2: Block Encoding and Linear Combination of Unitaries.

Aug 2022 — Designed and taught 3 lectures for a refresher module at **IIIT-Delhi**.
Topic: C programming for Operating Systems.

Teaching Assistantship (IIITD)

Theory of Computation — Winter 2020, Winter 2021, Winter 2024. **[Head TA]**

Data Structures — Summer 2022. **[Head TA]**

Intro to Quantum Computing — Winter 2023. **[Sole TA]**

Modern Algorithm Design — Monsoon 2020, Monsoon 2021. **[Sole TA]**

My duties included designing and conducting tutorials, creating assignments, and holding office hours and remedial sessions for all the above courses.

Research Visits and Internships

Aug 2024– Sep 2024 — **ACMU, Indian Statistical Institute**, Kolkata.
Host: Prof. Sourav Chakraborty.

Jul 2023– Aug 2023 — **Czech Technical University**, Prague.
Host: Prof. Jakub Marecek, Prof. Vyacheslav Kungurtsev.

Feb 2017 – Apr 2017 — **Systems Engineering Intern**, Infosys Limited.

Jun 2016– Jul 2016 — **Data-Science Intern**, AlCircle Pte Ltd.

Invited Talks

Oct 2025 — STCS seminar, TIFR Mumbai.
Generalization bounds for dependent data using online-to-batch conversion.

May 2025 — CS Colloquium, Ashoka University.
Efficient quantum agnostic improper learning of decision trees.

Mar 2025 — Quantum Computing Workshop by Edunautic, IIT Delhi.
Full day workshop from fundamentals to hands-on instruction.

Jan 2025 — Young Scientists Session at QAC 2025 symposium, C-DAC and DIAT Pune.
Modern Algorithmic Primitives in Quantum Computing.

Oct 2024 — ACMU seminar, ISI Kolkata.
Generalization bounds for dependent data using online-to-batch conversion.

July 2024 — Recent Trends in Algorithms Workshop 2024.
Efficient quantum agnostic improper learning of decision trees.

June 2024 — ACMU seminar, ISI Kolkata.
Efficient quantum agnostic improper learning of decision trees.

Februrary 2024 — Quantum Computing Semester, Chennai Mathematical Institute.
Quantum Algorithms for Linear Algebra.
Block Encodings and Linear Combination of Unitaries.

September 2023 — Center for Quantum Computing Science, University of Latvia.
Efficient quantum agnostic improper learning of decision trees.

Invited Talks (continued)

- IDA Seminar, Czech Technical University.
  Efficient quantum agnostic improper learning of decision trees.
- November 2022 — Short Talk at the 6th Quantum Techniques in Machine Learning (QTML 2022).
  Quantum boosting using domain-partitioning hypotheses.
- March 2022 — Theory Seminar, Indian Institute of Information Technology Delhi.
  Quantum boosting using domain-partitioning hypotheses.
- December 2020 — Faculty Development Programme, JNTU Anantapur.
  Quantum Machine Learning.

Miscellaneous

Awards and Achievements

- Feb 2026 — Invited poster at **ACM ARCS 2026**.
- Feb 2025 — Invited poster at **ACM ARCS 2025**. (Declined)
- Aug 2024 — **IIITD Dean's List** for Best Teaching Assistant (Theory of Computation).
- Apr 2024 — AISTATS 2024 Registration Grant.
- Jan 2022 — QIP 2022 Student Travel Award. (Not availed due to COVID restrictions)
- Feb 2020 — **Runners Up, IBMQ Awards** in the *Teach Me Quantum 2019* category. (4000 USD cash prize)
- Oct 2019 — Full Scholarship to attend the 4th Winter School in CSE organized by the IIAS, HUJI, Jerusalem.

Reviewing

- Conference — **STACS** ('25); **ICML** ('25,'26); **NeurIPS** ('24, '25); **ICLR** ('25,'26); **AISTATS** ('25,'26).
- Journal — **Scientific Reports** ('24); **Quantum** ('24, '25).

Mentoring

- Jan 2024 – May 2025 — Alhad Sethi. B.Tech student at IIIT-Delhi.
- Jan 2021 – Jul 2023 — Parmeet Singh Chani. B.Tech student at DTU, Delhi.
- Rohan Bhatia. B.Tech student at DTU, Delhi.

Workshop

- Workshops — Co-organiser and Webadmin for the QISE workshop at FSTTCS 2021.
- Seminars (IIIT-D) — Talks on Quantum Computing, Spectral Graph Theory, Theory Reading Group, Ketchup talks.

References

Available on Request